

Functional Specification Document (FSD)

Project: Gradebook Revision

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1. Overview

This specification defines the functional and behavioral requirements for the Gradebook Revision project. The goal is to provide teachers with greater control and efficiency in grading workflows through category-based weighting, pacing integration, visual grading indicators, and enhanced navigation and filtering.

2. User Flow Summary

2.1 Assignment Setup Flow 1. Teacher creates a new assignment. 2. The assignment form includes a new Category drop-down menu with options: Assignment (default), Lab, Speaking & Listening Exercise, Fitness Log, TDI Attendance, Test Prep Attendance, Extended Assignment. 3. If the teacher does not make a selection, the category remains Assignment. 4. Saved assignments display their selected category in the gradebook.

2.2 Teacher Gradebook Setup Flow 1. When a teacher opens the gradebook for a course for the first time, they are prompted with a Gradebook Setup window. 2. The window lists all potential grading categories with empty numeric input fields. 3. Teacher enters percentage weights into any number of categories. 4. Validation rule: total of all entered weights must equal 100%. - If not, an alert appears: "Weights must total 100% before saving." 5. Once saved, the configuration applies to all sections of the selected course unless the teacher specifies a section override. 6. Weights can be edited at any time; recalculations occur immediately after saving.

2.3 Grading Workflow Flow 1. Each student/assignment cell displays points earned / points possible. 2. The percentage column is calculated using the weighted formula if weights exist; otherwise, it uses simple points-earned logic. 3. Clicking a cell opens the student's submission directly. 4. Teacher can input grades manually, use keyboard arrows to navigate horizontally or vertically, or apply bulk actions (e.g., autofill entire column).

3. System Behavior & Logic

Category Dropdown: Appears in assignment setup. Default = "Assignment." Stored as assignment metadata. Grade Weights: Defined at section level. Used to calculate final grade percentages. Validation: total must equal 100%. Calculation Fallback: If no weights exist, use total points earned ÷ total points possible. Filtering: Teachers can view all students across sections or select a specific section. Search: Teachers can search by Student Name or Assignment Title. Navigation: Arrow keys move right (next assignment) or down (next student) and wrap at row/column end. Auto-Fill: Right-click or menu option allows entire column auto-fill (confirm modal before applying). Undo Exempt: Toggle "E" on/off; reverting restores grading availability and recalculates grades. Visual Indicators: New submissions = Red; Resubmissions = Yellow; Graded = Neutral. Pacing Guide Integration: When a due date passes with no submission, system applies a zero until submission + grading.

4. UI Elements

Gradebook Header Elements: - Search Bar: Searches by student name or assignment title. - Filter Menu: Dropdown: Teacher → Course → Section (optional). - Auto-Fill Button: Bulk apply value or exempt flag. - Column Navigation Arrows: Move between students or assignments.

4.2 Cell States

Ungraded (New Submission) - Red: Requires initial grading. Resubmission - Yellow: Indicates student resubmitted work. Graded - Neutral: Grading complete. Exempt - Gray 'E': Excluded from grade calculations.

5. Error Handling

Invalid Weight Total: Weights $\neq$ 100% → "Weights must total 100% before saving." Missing Category: Defaults to "Assignment." Missing Pacing Data: Display: "Pacing data unavailable for this course." Undo Exempt: Recalculate grade after confirmation.

6. Data Handling

Data Objects: - Assignment Categories: List of possible categories (static configuration). - Grade Weights: Teacher-defined numeric values (user input). - Student Submission: Assignment score, status (LMS database). - Pacing Guide: Assignment deadlines per student (pacing module).

7. Dependencies & Integration Points

Integrates with: Assignment module, Pacing guide module, Grading service, UI navigation component.

8. Acceptance Criteria Summary

Category selection available during assignment setup and defaults correctly. Weight setup must sum to 100% and persist per section. Calculation logic: Weighted average or fallback to points-based. Pacing zero logic applies after missed due date. Resubmission indicator displays correct color state and clears on grading. Filtering by course/section functions correctly. Navigation: Arrow keys and auto-fill operate as expected.

9. Future Enhancements

- Import/export grade weight templates - Multi-course configuration dashboard - Optional color customization for grading states - Analytics layer for tracking pacing compliance